

Clock Time Is Not the Double-Hour of Your Birth

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Abstract

A birth certificate records clock time. The double-hour of birth is something else. This note works through the difference between true solar time and mean solar time, and the three corrections that lie between a recorded clock time and the double-hour it must be read against. The first is daylight saving time: China observed it for six years, 1986 to 1991, and a time recorded in that window must be put back an hour. The second is longitude: Beijing time is the mean solar time of 120°E, not the local time of Beijing, and since China spans some sixty-two degrees of longitude under a single standard line, the correction reaches more than two hours in the west; the rule is one degree to four minutes, applied as $4 \text{ minutes} \times (\text{local longitude} - 120^\circ)$. The third is the equation of time, at most about sixteen minutes fast and fourteen minutes slow, which matters only when a birth time sits close to a double-hour boundary; two of its causes are the elliptical orbit and the obliquity of the ecliptic, and it passes through zero four times a year. The note then sets out the long-standing dispute over whether the 子 hour, which spans 23:00 to 01:00 and two calendar days, should turn the day at its initial (子初) or at its midpoint (子正), citing the *Gexiang xinshu* and the *Sanming tonghui*, and declines to choose between them. It closes with a four-step checklist: establish the place of birth, check daylight saving, correct for longitude, and apply the equation of time only on a boundary. The note is descriptive; it contains no predictive claims.

Keywords: true solar time; mean solar time; Beijing time; longitude correction; daylight saving time; equation of time; double-hour; 子 hour

1. Two rulers

True solar time: taken from the sun's actual position. The moment the sun reaches the true south and rises to its highest is the local noon. This is what the sundial measures.

Mean solar time: an imaginary second sun, moving uniformly along the celestial equator, taking exactly 24 hours every day of the 365-day year. This is what the clock keeps.

Why imagine one? Because the real one does not move uniformly.

The earth's orbit is an ellipse; by Kepler's laws it runs fast through perihelion in early January and slow through aphelion in early July. The sun's speed across the sky changes throughout the year, and a clock regulated by it would make the days unliveable.

So mean solar time is **uniform** and true solar time is **not**. Two rulers, measuring different things.

The ten fifty-six at Chengdu in the opening is a figure after the longitude correction, **with the equation of time not yet applied**. The equation of time is at most some tens of minutes; the fifth section of this note deals with it — when it must be reckoned and when it may be left aside.

2. Beijing time is not the time of Beijing

This is the layer most easily confused.

Beijing time is the mean solar time of the meridian 120°E — not the local time of Beijing. Beijing actually lies at about 116.4°E, so when the clock points to twelve, Beijing is still about 14 minutes short of noon.

More to the point: our country spans some 62 degrees of longitude from east to west (73°E to 135°E), and uses the single line of 120°E for the whole territory.

The standard time of a time zone is only an artificial time that everyone agrees to keep in common; it is not the local time of every place within that zone.

History has tried otherwise. In the Republican period the country was at one time divided into five time zones:

Time zone	Central meridian
Central Plains	120°E
Longshu	105°E
Xin-Zang	90°E
Kunlun	82.5°E (some say 75°)
Changbai	127.5°E (some say 135°)

Later they were unified into one. Unification has its advantages; the price is that clock time and local solar time fall out of step.

3. One degree, four minutes

The earth turns 360 degrees in 24 hours, so 1 degree = 4 minutes.

The formula is simple:

真太阳时 = 北京时间 + 4 分 × (当地经度 - 120°) + 均时差 (True solar time = Beijing time + 4 minutes × (local longitude - 120°) + the equation of time.)

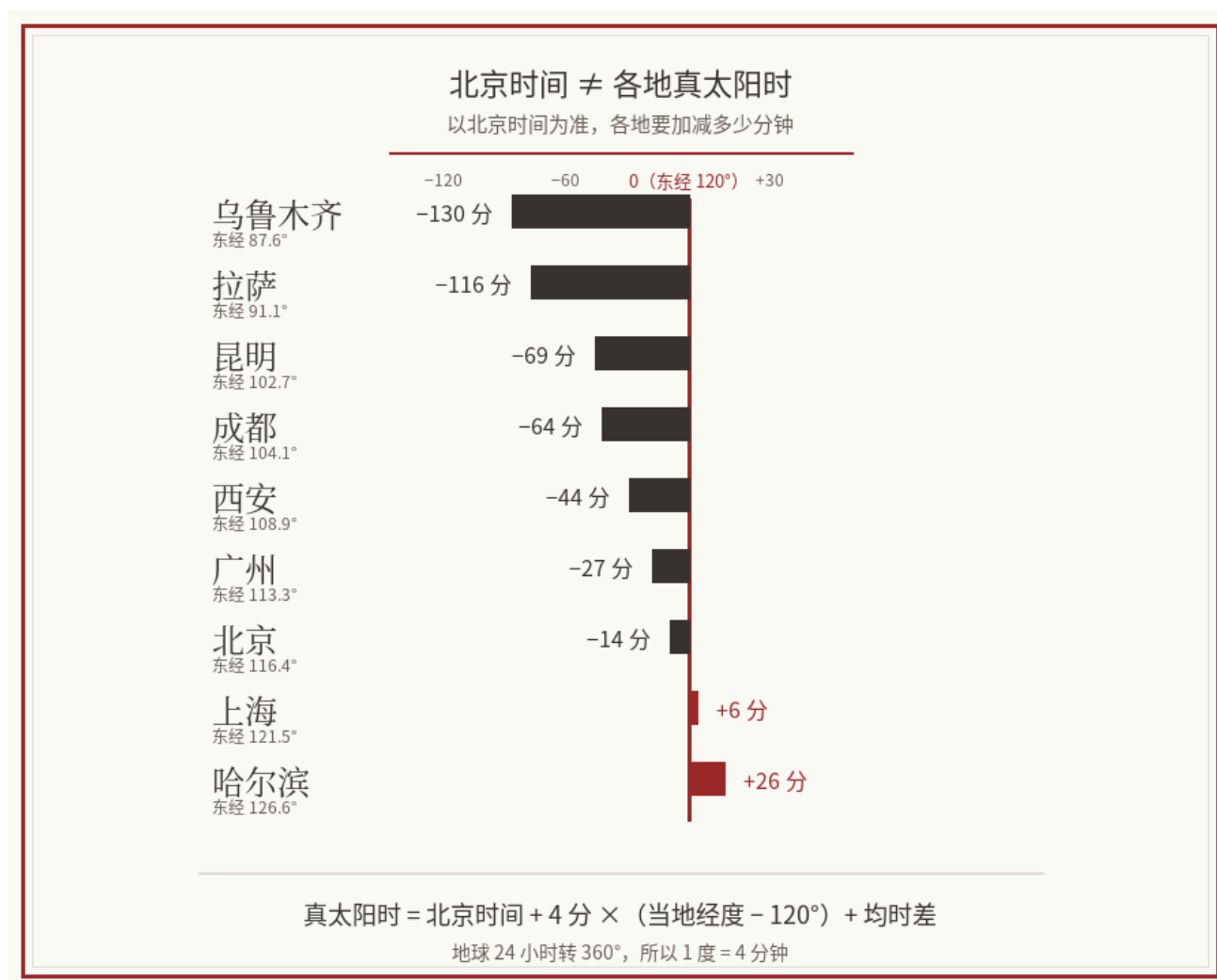
Longitude corrections for the main cities:

City	Longitude	Correction
Ürümqi	87.6°	-130 min
Lhasa	91.1°	-116 min
Kunming	102.7°	-69 min
Chengdu	104.1°	-64 min
Xi'an	108.9°	-44 min
Guangzhou	113.3°	-27 min
Shenzhen	114.1°	-24 min
Beijing	116.4°	-14 min
Shanghai	121.5°	+6 min
Harbin	126.6°	+26 min

Two examples:

Shenzhen, born at 9:10 Beijing time. Correction -24 min → true solar time 8:46. 9:10 is the 巳 hour, 8:46 is the 辰 hour. One double-hour apart.

Shanghai, born at 23:55 Beijing time. Correction +6 min → true solar time 0:01 the next day. It crosses zero o'clock, and the day-pillar changes.



4. The cut of daylight saving

There is a further error that is artificial, and the most easily overlooked.

From 1986 to 1991 the whole country observed daylight saving for six years: from 2 a.m. on the first Sunday in mid-April the clock was put forward one hour, and from 2 a.m. on the first Sunday in mid-September it was put back.

- 4 May — 14 September 1986 (the first year was special)
- 12 April — 13 September 1987
- 10 April — 11 September 1988
- 16 April — 17 September 1989
- 15 April — 16 September 1990
- 14 April — 15 September 1991

From 5 April 1992 it was suspended.

For those born in these six years, the time recorded on the birth certificate is the time after the clock was put forward, and one hour must be subtracted to restore it. Many people do not know they were once "put forward", and cast their charts straight away, wrong at the first step.

5. The equation of time: this layer can be left to the last

The difference obtained by subtracting mean solar time from true solar time is the equation of time.

- Over the year: at most about 16 minutes fast and about 14 minutes slow
- It is zero four times in a year
- Two causes: the elliptical orbit (about ± 7.66 min) and the obliquity of the ecliptic (about ± 9.87 min)

Set the three levels of error side by side, and their weights are very far apart:

Level	Source of error	Maximum	Must it be reckoned?
First	daylight saving time	± 60 min	Yes
Second	longitude	up to more than 2 hours	Yes
Third	equation of time	± 16 min	only when it sits on a double-hour boundary

That is: for most people the longitude correction is enough. The equation of time needs to be brought out only when the birth time lies exactly on the boundary between double-hours, and within twenty minutes of it either way.

Do not give the impression that one "must reckon to the second" — that is bluffing.

6. The 子 hour: how those two hours are reckoned

The 子 hour runs 23:00–01:00 and spans two days. Here there is an old question of a thousand years.

School	Position	Basis
Day changes at 子初	23:00 already counts as the new day	the Yuanhai ziping, the method of "raising the double-hour from the day"; folk tradition; "the double-hour may not be divided"
Day changes at 子正	00:00 is the boundary. For a midnight 子 hour the day-pillar takes the present day and the hour-pillar takes the 子 hour	the Sanming tonghui, "On the Double-Hours"; the Xin Tangshu, "Treatise on the Calendar", "古历分日，起于子半"

The clearest statement in the sources is Zhao Youqin of the Yuan (courtesy name Yuandu), in the Gexiang xinshu:

若子时则上半时在夜半前属昨日，下半时在夜半后属今日。(In the case of the 子 hour, the first half falls before midnight and belongs to the previous day; the second half falls after midnight and belongs to the present day.)

Wan Mingyong of the Ming, in the Sanming tonghui, "On the Double-Hours", has a passage identical to it word for word:

子时则上半时在夜半前，属昨日；下半时在夜半后，属今日。(The 子 hour: the first half falls before midnight and belongs to the previous day; the second half falls after midnight and belongs to the present day.)

Each view has its own textual lineage; this note makes no recommendation — take whichever your own transmission prescribes.

Hence the folk saying, "神仙难断子时命 — not even the immortals can settle the fate of a 子-hour birth".

But more important than the dispute is the order:

先扣夏令时 → 再按经度换成真太阳时 → 看是否落进子时区间 → 最后才谈早晚子时 (First deduct daylight saving → then convert by longitude to true solar time → see whether it falls within the 子-hour interval → only then speak of the early or late 子 hour.)

Without the longitude correction first, whether it is even the 子 hour is not yet settled, and arguing about the early or late 子 hour is meaningless.

7. A four-step checklist

From birth time to the hour-pillar, there are just these four steps:

1. Establish the place of birth — not "where are you from" but "at which hospital". A province's difference may be a double-hour's difference.
2. Check daylight saving — for anyone born in 1986–1991, subtract one hour first.
3. Apply the longitude correction — formula: Beijing time + 4 min $\times$ (local longitude – 120°).
4. Apply the equation of time only on a boundary — if after the first two steps the time is still more than half an hour from the boundary, it may be ignored.

Closing

Back to the title: the clock time of birth is not the double-hour of birth.

Where is the difference? One comes from the sun, one from the clock, and one is a hand pulling the clock forward — and what casting a chart needs is the one from the sun.

The clock is unified across the country; the sun is not.

When the double-hours were discussed earlier, it was said that the ancients measured them with a sundial — measured, that is, the sun. So we are not inventing any new rule, but recovering the same ruler.

What comes next is the stems and branches — the ten heavenly stems and the twelve earthly branches: what those twenty-two signs actually are.

Declarations

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Data availability. No new data were created or analysed in this study.

Language. The original Chinese text is deposited as a companion file to this record (钟表上的出生时间，不等于你出生的时辰).

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